# PHYSICS

1) A uniform rod of length 1.0 metre is bent at its midpoint to make 90° (Ninty degree). The distance of the centre of mass from the centre of the rod is :-

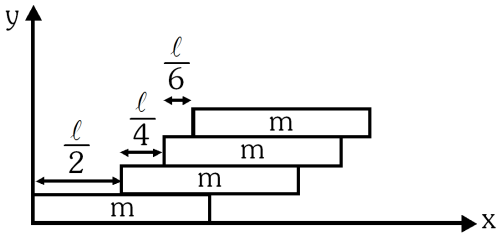
(1) $\frac{1}{\sqrt{2}}$m

(2) $\frac{1}{2\sqrt{2}}$ m

(3) $\frac{1}{4\sqrt{2}}$ m

(4) $\sqrt{2}$m

2) Find the x coordinate of the centre of mass of the bricks shown in figure :



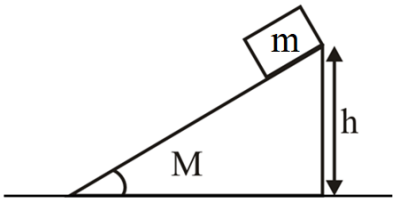

(1) $\frac{24}{25}\ell$

(2) $\frac{25}{24}\ell$

(3) $\frac{15}{16}\ell$

(4) $\frac{16}{15}\ell$

3) A mass m rest on an inclined plane of mass M which is further resting on a smooth horizontal plane. Now if the mass starts moving the position of centre of mass of the system will :-




(1) Remain at rest

(2) Move in horizontal direction

(3) Move vertically upward

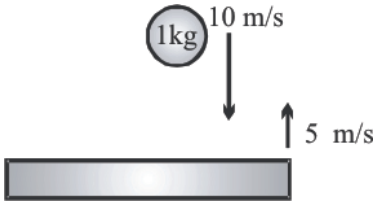
(4) Move vertically downward

4) A 50 gm bullet moving with a velocity of 10 m/s gets embedded into a 950 gm stationary body.

The loss in kinetic energy of the system will be –

(1) 5%
(2) 50%
(3) 100%
(4) 95%

5) A ball of mass 1 kg strikes a heavy platform, elastically, moving upwards with a velocity of 5m/s. The speed of the ball just before the collision is 10 m/s downwards. Then the impulse imparted by the platform on the ball is

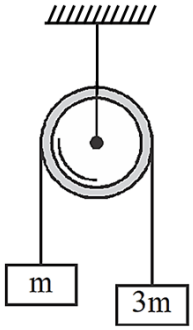



(1) 15 N–s
(2) 10 N–s
(3) 20 N–s
(4) 30 N–s

6) A ball of mass 10 kg is moving with a velocity of 10 m/s. It strikes another ball of mass 5 kg which is moving in the same direction with a velocity of 4 m/s. If the collision is elastic, their velocities after the collision will be, respectively

(1) 6 m/s, 12 m/s
(2) 12 m/s, 6 m/s
(3) 12 m/s, 10 m/s
(4) 12 m/s, 25 m/s

7) If the system is released, then the acceleration of the centre of mass of the system :




(1) $\frac{g}{4}$
(2) $\frac{g}{2}$
(3) g
(4) 2g

8) A cannon ball is fired with a velocity 200 m/sec at an angle of 60° with the horizontal. At the highest point of its flight it explodes into 3 equal fragments, one going vertically upwards with a

velocity 100 m/sec, the second one falling vertically downwards with a velocity 100 m/sec. The third fragment will be moving with a velocity

(1) 100 m/s in the horizontal direction

(2) 300 m/s in the horizontal direction

(3) 300 m/s in a direction making an angle of 60° with the horizontal

(4) 200 m/s in a direction making an angle of 60° with the horizontal

9) A cart of mass M is tied to one end of a massless rope of length 10m. The other end of the rope is in the hands of a man of mass $\frac{M}{2}$. The entire system is one a smooth horizontal surface. If the man pulls the cart by the rope then find the distance the man will slip on the horizontal surface before the man & the cart will meet each other.

(1) $\frac{20}{3}$m

(2) $\frac{10}{3}$m

(3) 0

(4) None

10) A ball of mass m falls vertically to the ground from a height $h_1$ and rebound to a height $h_2$. The change in momentum of the ball striking the ground is :-

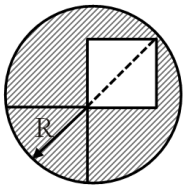
(1) $mg(h_1 - h_2)$

(2) $m\left[\sqrt{2gh_1} + \sqrt{2gh_2}\right]$

(3) $m\left[\sqrt{2g(h_1 + h_2)}\right]$

(4) $m\left[\sqrt{2g}(h_1 + h_2)\right]$

11) From a circular disc of radius R, a square is cut out with a radius as its diagonal. The center of mass of remainder part is at a distance (from the centre):-



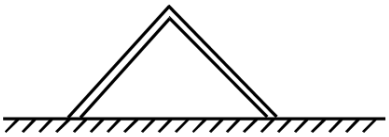

(1) $\frac{R}{(4\pi - 2)}$

(2) $\frac{R}{2\pi}$

(3) $\frac{R}{(\pi - 2)}$

(4) $\frac{R}{(2\pi - 2)}$

12) Two equal rods joined at one end are kept on a smooth surface as shown and released.

Trajectory of centre of mass of both rods is - 

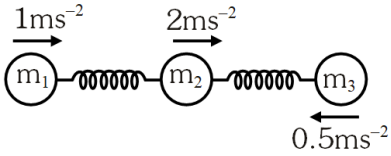
(1) parabola

(2) straight vertical line

(3) straight inclined line

(4) straight horizontal line

13) Shown in the figure is a system of three particles having masses $m_1$ = 1 kg, $m_2$ = 2 kg and $m_3$ = 4kg connected by two springs. At an instant, accelerations of these particles are 1 m/s$^2$, 2 m/s$^2$ and 0.5 m/s$^2$ respectively as shown? The external force at this instant acting on the system is :



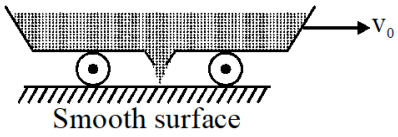

(1) 1 N rightward

(2) 3 N leftward

(3) 3 N rightward

(4) Zero

14) A trolley filled with sand moves on a smooth horizontal surface with a velocity v0. A small hole is made at the base of it from which sand is leaking out at constant rate. As the sand leaks out then which of the following is **incorrect**?




(1) The velocity of the trolley increases

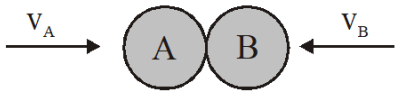
(2) The velocity of the trolley remains constant

(3) The momentum of the trolley is conserved

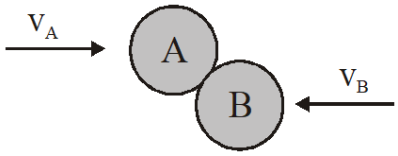
(4) The momentum of the total system (trolley with sand + leaked out sand) is conserved in horizontal direction

15) Two bodies, A and B, collide as shown in Figures a and b below.

(a)




(b)




Which statement is true?

(1) They exert equal and opposite forces on each other in (a) but not in (b).

(2) They exert equal and opposite force on each other in both (a) and (b).

(3) The forces are equal and opposite to each other in (a), but only the components of the forces parallel to the velocities are equal in (b).

(4) The forces are equal and opposite in (a), but only the components of the forces perpendicular to the velocities are equal in (b)

16) A coin is placed on the horizontal surface of a rotating disc. The distance of the coin from the axis is 1 m and coefficient of friction is 0.5. If the disc starts from rest and is given an angular acceleration $\frac{1}{\sqrt{2}}$ rad / sec$^2$, then the number of revolutions through which the disc turns before the coin slips is

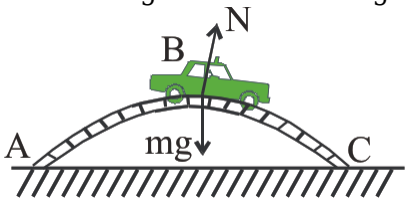
(1) $\frac{4}{\pi}$

(2) $\frac{7}{4\pi}$

(3) $\frac{5}{4\pi}$

(4) $\frac{3}{4\pi}$

17) A car is going on an overbridge of radius R, maintaining a constant speed. As the car is descending on the overbridge from point B to C, the normal force on it :




(1) increase

(2) decreases

(3) remains constant

(4) first increases then decreases.

18) ***Assertion (A):*** During a turn, the value of centripetal force should not be more than the limiting frictional force.
***Reason (R):*** The centripetal force is provided by the frictional force between the tyres and the road.

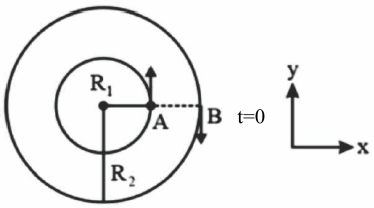
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

19) In the figure given, A and B are two particles having same magnitude of angular velocity in opposite sense, Then $\vec{V_A} - \vec{V_B}$ at $t = \pi/2\omega$ sec is



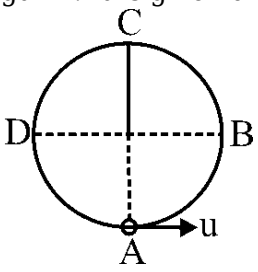
(1) $-\omega R_1\hat{i} + \omega R_2\hat{j}$

(2) $\omega R_1\hat{i} - \omega R_2\hat{j}$

(3) $\omega R_2\hat{i} - \omega R_1\hat{j}$

(4) $\omega R_1\hat{i} + \omega R_2\hat{j}$

20) A particle is suspended from a string of length R. It is given a velocity u = $3\sqrt{gR}$ at the lowest point, then match the column-I with column-II.




| Column-I | | Column-II | |
|---|---|---|---|
| (P) | Velocity at B | (1) | 7 mg |
| (Q) | Velocity at C | (2) | $\sqrt{5gR}$ |
| (R) | Tension in string at B | (3) | $\sqrt{7gR}$ |
| (S) | Tension in string at C | (4) | 4 mg |

(1) P→3; Q→1; R→2; S→4

(2) P→3; Q→2; R→1; S→4

(3) P→4; Q→2; R→1; S→3

(4) P→1; Q→4; R→2; S→3

21) A particle of mass m starts moving in a circular path of constant radius r, such that its centripetal acceleartion $a_c$ is varying with time t as $a_c = k^2rt^2$, where k is a constant. What is the power delivered to the particle by the force acting on it :

(1) $mkr^2t$

(2) $m^2k^2r^2t^2$

(3) $mk^2r^2t$

(4) None of these

22) A particle of mass m is fixed to one end of a light spring of force constant k and unstretched length  The system is rotated about the other end of the spring with an angular velocity ω, in gravity free space. Then increase in length of the spring will be:

(1) $\frac{m\omega^2\ell}{k}$

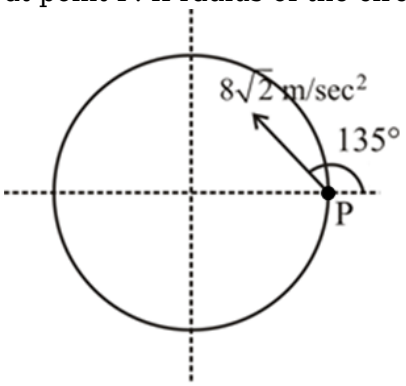
(2) $\frac{m\omega^2\ell}{k - m\omega^2}$

(3) $\frac{m\omega^2\ell}{k + m\omega^2}$

(4) None of these

23) The acceleration of the particle performing non uniform circular motion is shown in the diagram

at point P. If radius of the circular path is 2m then what is the velocity at point P.




(1) 4 m/sec

(2) 2 m/sec

(3) $3\sqrt{2}$ m/sec

(4) 6 m/sec

24) A particle moves so that its position vector is given by $\vec{r} = \cos\omega t\hat{i} + \sin\omega t\hat{j}$. Where ω is a constant. Which of the following is true ?

(1) Velocity and acceleration both are perpendicular to $\vec{r}$.

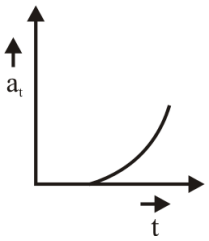
(2) Velocity and acceleration both are parallel to $\vec{r}$

(3) Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin

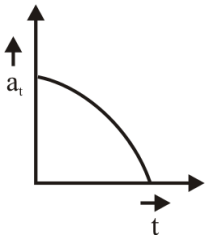
(4) Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin

25) The angle turned by a body under going circular motion depend on time as $\theta = \theta_0 + \theta_1 t + \theta_2 t^2$ then draw tangent acceleration $a_t$ v/s time t graph for the body. (Radius = constant).

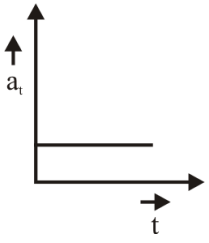
(1)



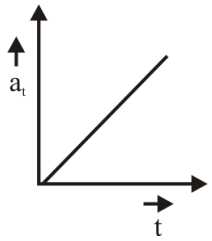

(2)



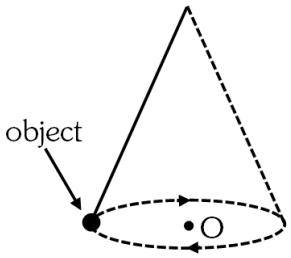

(3)



(4)




26) A small object, suspended by a string, rotates with constant speed, in a horizontal circle as shown in the figure. Point O is at the centre of the circle. Neglect air drag.




Which one of the following statements is correct?

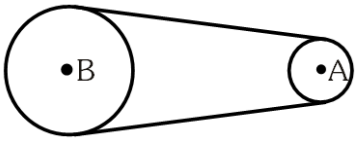
(1) The object is in equilibrium.

(2) There is a resultant force on the object directed away from O.

(3) A force acts on the object in the direction of its motion.

(4) There is a resultant force on the object directed towards O.

27) A wheel of radius 0.1 m (wheel A) is attached by a non stretching belt to a wheel of radius 0.2 m (wheel B). The belt does not slip. By the time, wheel B turns through one revolution, wheel A will rotate through x revolution. Find the value of x.



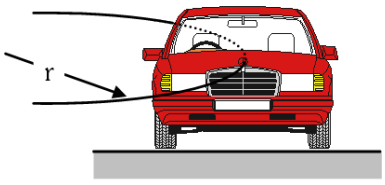

(1) 6

(2) 4

(3) 2

(4) 0

28) Two-wheeler can tilt while turning, but four-wheeler cannot. Estimate maximum safe speed of a four-wheeler on turn of radius 10 m assuming coefficient of static friction between tires and the flat road to be 0.64 and acceleration due to gravity to be 10 $m/s^2$.




(1) 6.4 m/s

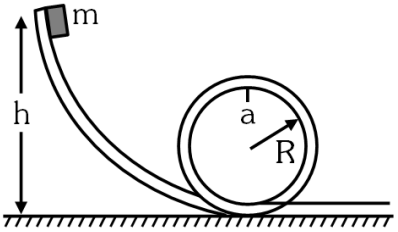
(2) 2.8 m/s

(3) 8 m/s

(4) 10 m/s

29) An object of mass m is released from rest at a height h above the surface of a table. The object

slides along the inside of the loop. The loop track consisting of a ramp and a circular loop of radius R shown in the figure. Assume that the track is frictionless. When the object is at the top of the circular track it pushes against the track with a force equal to three times its weight. What height was the object dropped from ?



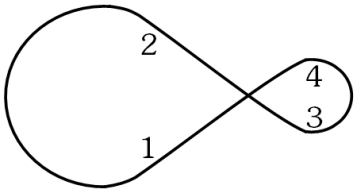

(1) 3R

(2) 4R

(3) 5R

(4) 6R

30) A car runs at constant speed around the horizontal race track shown in the figure. Over which portion of the track is the magnitude of the acceleration the greatest?



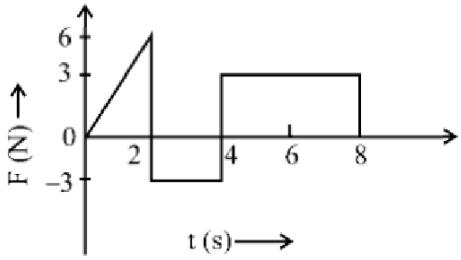

(1) From 1 to 2

(2) From 2 to 3

(3) From 3 to 4

(4) From 4 to 1

31) The force 'F' acting on a particle of mass 'm' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :-




(1) 24 Ns

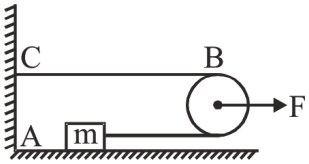
(2) 20 Ns

(3) 12 Ns

(4) 6 Ns

32)

The acceleration of massless pulley is :-



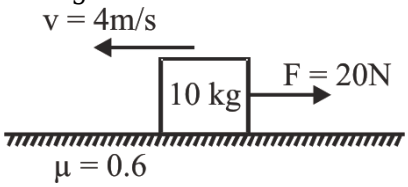

(1) F/m

(2) F/2m

(3) F/4m

(4) F/8m

33) A block of mass 10 kg is moving on a rough surface as shown in figure. The frictional force acting on block is :-



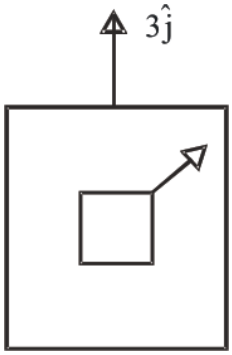

(1) 20 N

(2) 60 N

(3) 40 N

(4) 80 N

34) A block is placed on a rough plank moving with velocity $3\hat{j}$ and velocity of block relative to plank is $3\hat{i} + 4\hat{j}$, find unit vector in direction of kinetic friction force acting on the block is :-



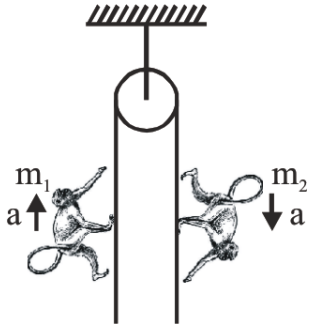

(1) $\frac{-(3\hat{i} + \hat{j})}{\sqrt{10}}$

(2) $\frac{-(3\hat{i} + 4\hat{j})}{5}$

(3) $\frac{(3\hat{i} + 4\hat{j})}{5}$

(4) $\frac{3\hat{i} + \hat{j}}{\sqrt{10}}$

35) A monkey climbs up and another monkey moving down a rope passing over pulley with same uniform acceleration separately. If the respective masses of monkeys are in the ratio 2 : 3, the




common acceleration must be :-

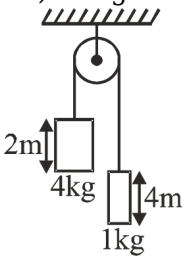
(1) $\frac{g}{5}$

(2) 6g

(3) $\frac{g}{2}$

(4) g

36) In figure shown, both blocks are released from rest. Find the time to cross each other ?




(1) 1 sec

(2) $\sqrt{2}$ sec

(3) 2 sec

(4) $\frac{1}{\sqrt{2}}$ sec

37) The exhaust velocity of gases with respect to a small rocket of mass 25 kg is $(49 \times 10^2)$ m/s. At what rate the fuel must burn so that it may rise up with an acceleration of 9.8 $m/s^2$ :-

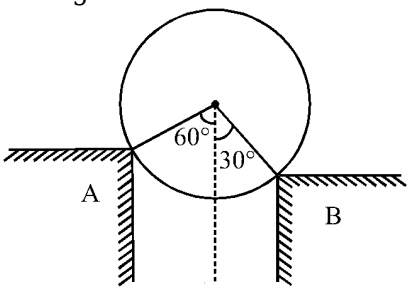
(1) 0.5 kg/s

(2) 0.1 kg/s

(3) 1.5 kg/s

(4) 0.15 kg/s

38) A smooth cylinder of mass m and radius R is resting on two corner edges A and B as shown in fig. The relation between normal reaction at the edges A and B is :-



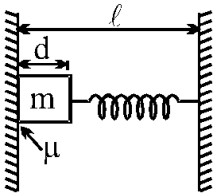

(1) $N_A = \sqrt{2}N_B$

(2) $N_B = \frac{2\sqrt{3}N_A}{5}$

(3) $N_A = \frac{N_B}{2}$

(4) $N_B = \sqrt{3}N_A$

39) A block of mass m is pressed against a vertical surface by a spring of unstretched length □. If the coefficient of friction between the block and the surface is μ. Choose the correct statement.



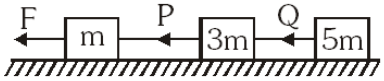

(1) if spring constant k = $\frac{2mg}{\mu d}$, block will not be in equilibrium.

(2) if spring constant is k = $\frac{2mg}{\mu d}$, the normal reaction is $\frac{mg}{\mu}$

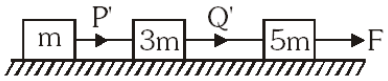
(3) in the part (2) force of friction is 2mg

(4) minimum spring constant $k_{min}$ to keep the block of mass m in equilibrium is $\frac{mg}{\mu d}$

40) Three blocks of masses m, 3m and 5m are connected by massless strings and pulled by a force F on a frictionless surface as shown in the figure below. The tension 'P' in the first string is 16N.




If the point of application of F is changed as given below, the values of tension P' and Q' shall be respectively :-



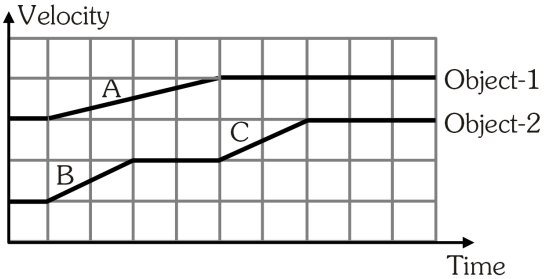

(1) 16N, 10N

(2) 10N, 16N

(3) 2N, 8 N

(4) None of these

41) The graph shows the velocities of two objects as function of time. Mass of object-1 is twice of that of object-2. Net forces $F_A$, $F_B$, and $F_C$ are acting on the objects during intervals A, B, and C, respectively. Which one of the following choices is the correct relationship between the magnitudes of the net forces?



(1) $F_B = F_C > F_A$

(2) $F_C = F_B = F_A$

(3) $F_A > F_B = F_C$

(4) $F_A > F_B > F_C$

42) Among the given cases, in which cases net force on the object will be zero ?
(a) An ice cube of 50g mass floating on water
(b) A body moving with uniform velocity in space
(c) A satellite revolving around the earth
(d) A book at rest on a table

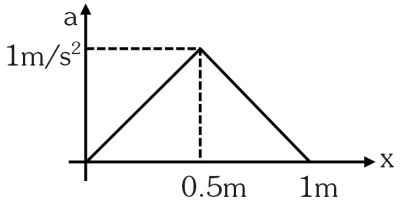
(1) In a, b and c

(2) only in a

(3) In a, b and d

(4) in all cases

43) A body initially at rest, starts moving along x-axis in such a way so that its acceleration vs displacement plot is as shown in figure. The maximum velocity of particle is :-



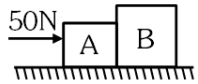

(1) 1 m/s

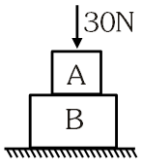
(2) 6 m/s

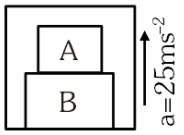
(3) 2 m/s

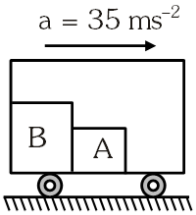
(4) none

44) In which of the following cases the contact force between A & B is maximum : ($m_A = m_B = 1kg$)

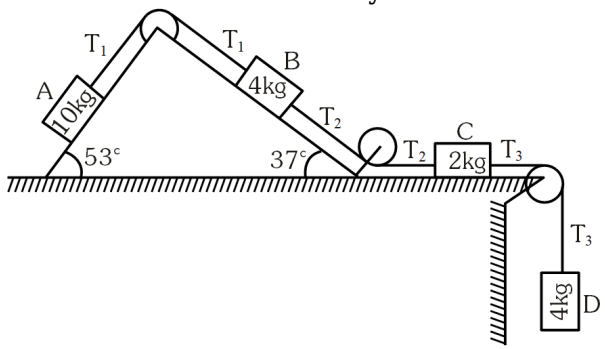
(1) 


(2) 


(3) 


(4) 

45) In shown system all surfaces are frictionless. All strings are massless and all pulleys are frictionless and massless. System is released from rest from shown position. [g = 10 m/s$^2$]




Acceleration of block C is

(1) 0.6 m/s$^2$ towards right

(2) 0.8 m/s$^2$ towards left

(3) 0.8 m/s$^2$ towards right

(4) $\frac{8}{9}$ m/s$^2$ towards left

## CHEMISTRY

1) Select the correct statement :-
(a) $Mn^{+3}$ act as stronger oxidising agent than $Cr^{+3}$ & $Fe^{+3}$
(b) $MnF_7$, $SiF_6^{-2}$ & $BF_6^{-3}$ all are known flouride
(c) $La^{+3}$ & $Lu^{+3}$ both are diamagnetic
(d) When $K_2Cr_2O_7$ react with $H_2O_2$ in acidic medium it forms dark orange ($CrO_5$) compound

(1) a,b,d

(2) a,b,c

(3) a,c

(4) a,b,c,d

2) Which of the following statement is not true ?

(1) $K_2Cr_2O_7$ solution in acidic medium is orange

(2) $K_2Cr_2O_7$ solution becomes yellow on increasing the pH beyond 7

(3) On passing $H_2S$ through acidified $K_2Cr_2O_7$ solution, a milky colour is observed

(4) The bond angle of Cr-O-Cr < O-Cr-O in $K_2Cr_2O_7$

3) Match List-I with List-II and identify correct option.

| | **List-I** | | **List-II** |
|---|---|---|---|
| (P) | Colourless in aqueous medium | (i) | $Cu^{2+}$, $Fe^{3+}$ |
| (Q) | μ = 5.9 BM | (ii) | $Ti^{+3}$, $Cu^{2+}$ |

| | | | |
|---|---|---|---|
| (R) | Coloured in aqueous medium | (iii) | $Fe^{3+}$, $Mn^{2+}$ |
| (S) | $\mu$ = 1.7 BM | (iv) | $Zn^{2+}$, $Sc^{3+}$ |

(1) (P) → (iii), Q → (iv), (R) → (i), S → (ii)

(2) (P) → (iv), Q → (iii), (R) → (i), S → (ii)

(3) (P) → (ii), Q → (iii), (R) → (iv), S → (i)

(4) (P) → (i), Q → (ii), (R) → (iii), S → (iv)

4) Which of the following reaction condition is correct for prepation of $KMnO_4$ / $NaOH^-$

(1) $MnO_2 \xrightarrow[2)\text{ Oxidised with air or } KNO_3]{1)\text{ Fused with KOH}} \xrightarrow{3)\text{ Electrolyticoxidation in alkaline solution}}$

(2) $\underset{\text{salt}}{Mn^{+2}} \xrightarrow{\text{Oxidised by peroxidisulphate}}$

(3) $MnO_2 \xrightarrow[2)\text{ Oxidised by air}]{1)\text{ Fused with } K_2CO_3 \text{ or KOH}} \xrightarrow{3)\text{ Disproportionates in neutral or acidic solution}}$

(4) All of these

5) When lanthanides heated with carbon which carbide is not formed ?

(1) $Ln_2C_3$

(2) $LnC_2$

(3) $Ln_3C$

(4) $Ln_2C_2$

6)

Which of the following statement is incorrect :

(1) $Pb^{+4}$ is a reducing agent while $Sn^{+4}$ is a oxidising agent

(2) Phosphorous acid show disproportionation reaction on heating

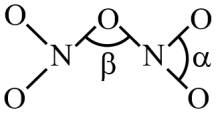
(3) $F_2$ is the only non metal that can oxidise $H_2O$

(4) $XeO_3$ is colourless explosive solid

7) Which of the following option is incorrect ?

(1) Melting point :- As > Sb > Bi > P > N

(2) Nitrogen has two stable isotopes $^{14}N$ & $^{15}N$

(3)  $\alpha = \beta$

(4) $2Se_2Cl_2 \rightarrow SeCl_4 + Se$

8) Find the number of molecules, when they undergo in hydrolysis at room temperature & the produced acid from central atom has the basicity of '2'.
($PCl_5$, $SF_6$, $SF_4$, $P_4O_6$, $PCl_3$)

(1) 4

(2) 3

(3) 2

(4) 5

9) Consider the following and mark (T) for true and (F) for false statement for group 16.
(i) $SF_4$ is a gas and $TeF_4$ is a liquid.
(ii) Among hexa halides hexa fluorides are the only stable halides.
(iii) The absence of d orbital oxygen limits in covalency to four and in practice, rarely exceeds two
(iv) Reducing property of dioxide increases from $SO_2$ to $TeO_2$.

(1) T T F T

(2) F F F T

(3) F T T F

(4) F F F F

10) **Statement- 1 ($S_1$) :** $Xe[PtF_6]$ & $O_2[PtF_6]$ are red colour compound having +5 oxidation state of Pt.
**Statement- 2 ($S_2$) :** First ionisation energy of molecular oxygen is almost similar with xenon.

(1) $S_1$ & $S_2$ both are correct

(2) $S_1$ & $S_2$ both are incorrect

(3) $S_1$ is correct & $S_2$ is incorrect

(4) $S_1$ is incorrect & $S_2$ in correct

11) Match the column :

| | | | |
|---|---|---|---|
| (A) | $ClO_2$ | (P) | Paramagnetic in nature |
| (B) | $Cl_2O_6$ | (Q) | Hydrolysis gives $HClO_3$ |
| (C) | $Cl_2O_7$ | (R) | Cl–O–Cl bond |
| (D) | $Cl_2O$ | (S) | Acidic oxide |
| | | (T) | $sp^3$ hybridisation (Central atom) |

(1) A→P,Q ; B→Q,S,T ; C→R,S ; D→S,T

(2) A→Q,S,T ; B→Q,S,T ; C→R,S ; D→S,T

(3) A→P,Q,S ; B→Q,S,T ; C→R,S,T ; D→S,T

(4) A→Q,S ; B→Q,S,T ; C→R,S,T ; D→Q,S

12) Characteristic bonds and their number in pyrophosphoric acid:

(1) Four P–OH, Two P = O, One P–O–P

(2) Four P–OH, one P = O, two P–O–P

(3) Four P–OH, one P = O, three P–O–P

(4) Four P–OH, Two P = O, three P–O–P

13) Which of the following is incorrect ?

(1) Pure ozone is a pale blue gas

(2) $O_3$ is thermodynamically unstable w.r.t. $O_2$

(3) Both Rhombic and monoclinic sulphur have same crystal structures

(4) Stability of oxides ⇒ I > Cl > Br

14) Which of the following elements react with steam

(1) C(s)

(2) Si(s)

(3) Pb(s)

(4) Sn

15) “Middle row anamoly” is related with

(1) stability of transition metal oxides

(2) stability of alkali metal oxides

(3) stability of halogen oxides

(4) stability of alkaline earth metal oxides

16) Mark the compound which is soluble in hot water.

(1) Lead chloride

(2) Mercurous chloride

(3) Strontium sulphate

(4) Silver chloride

17) Lime react exothermally with water to give 'A' which has low solubility in water. Aq. Sol of 'A' often used for least of $CO_2$, a test in which insoluble B is formed if B is further reacted with $CO_2$ then soluble compound in formed 'A' is :

(1) Quick lime

(2) Slaked lime

(3) Lime water

(4) White lime

18) If HCl is not added before passing $H_2S$ in the second group, it may result in the :

(1) incomplete precipitation of second group sulphide

(2) precipitation of sulphides of cations belonging to subsequent groups

(3) precipitation of sulphur

(4) precipitation of lead as lead sulphide

19) Incorrect match is :

| | Identification test | Colour |
|---|---|---|
| (1) | Starch - iodine test | Blue |
| (2) | Chromyl chloride test | Green |
| (3) | Layer test of bromide | Red-Brown |
| (4) | Layer test of Iodide | Violet |

(1) 1

(2) 2

(3) 3

(4) 4

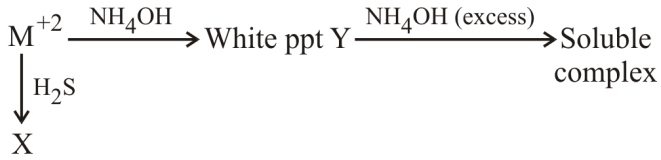
20) **Assertion :** Test of $NH_4^+$ cannot be done within group analysis.
**Reason :** During group analysis several times $NH_4^+$ compound is added at the different steps.

(1) Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.




21) (white ppt)

$M^{+2}$ is :-

(1) $Cu^{+2}$

(2) $Ni^{+2}$

(3) $Hg^{+2}$

(4) $Zn^{+2}$

22) If two substances A and B have $P_A° : P_B° = 1 : 2$ and have mole fraction in solution in the ratio 1 : 2 then mole fraction of A in vapour phase :-

(1) 0.33

(2) 0.25

(3) 0.20

(4) 0.52

23) What will be the freezing point of 0.2 molal aqueous solution of $MgBr_2$ ? If salt shows 60% dissociation in solution and $K_f$ for water is1.86 K Kg $mol^{-1}$.

(1) –0.59 °C

(2) –0.67 °C

(3) –0.82 °C

(4) –1.04 °C

24) What will be the molality of 2M $H_2SO_4$ solution having density 1.8 g/mL :-

(1) 3M

(2) 1.25 M

(3) 2.4 M

(4) 4.8 M

25) Which of the following have highest Boiling point

(1) 0.1 m Glucose

(2) 0.2 m $Al_2(SO_4)_3$ ($\alpha = 1$)

(3) 0.3 m Sucrose

(4) 0.4 m NaCl ($\alpha = 1$)

26) The value of $K_b$ is given by $R(T_b^o)^2/1000X$, in this relation X is :-

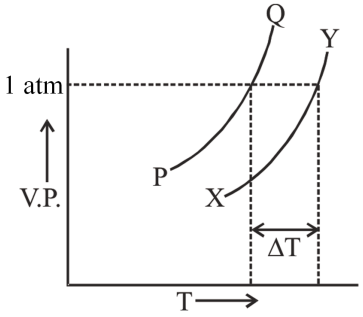
(1) Density of solution

(2) $\Delta H_f^o$ of solvent

(3) Latent heat of vaporization

(4) $\Delta H_{vap}$.

27) In the graph plotted between vapour pressure (V.P.) and temperature (T) :-




(1) PQ is the curve for solvent, XY is the curve of solution and ΔT is depression in freezing point

(2) PQ is the curve of solution, XY is the curve for solvent and ΔT is elevation in boiling point

(3) PQ is the curve for solvent, XY is the curve for solution and ΔT is molal elevation in Freezing point

(4) PQ is the curve for solvent, XY is the curve for solution and ΔT is elevation in boiling point.

28) Match List - I with List - II.

| List - I | List - II |
|---|---|

| | | | |
|---|---|---|---|
| (a) | 0.001M $Na_2SO_4$ and<br>0.003M glucose at same temperature | (i) | Hypotonic solution |
| (b) | 0.003M $Na_2SO_4$ is _____ with<br>0.003M glucose at same temperature | (ii) | iCRT |
| (c) | 0.001M $Na_2SO_4$ is _____ with<br>0.01M glucose at same temperature. | (iii) | Isotonic solution |
| (d) | Osmotic pressure | (iv) | Hypertonic solution |

(1) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

(2) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)

(3) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

(4) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

29) Incorrect thermodynamic relation for ideal solution is :

(1) $\Delta_{mix} G < 0$

(2) $\Delta_{mix} S < 0$

(3) $\Delta_{mix} H = 0$

(4) $\Delta_{mix} V = 0$

30) Which one has same van't hoff factor (i) as that of $Hg_2Cl_2$ ($\alpha = 1$)

(1) $AlCl_3$ ($\alpha = 1$)

(2) $CaCl_2$ ($\alpha = 1$)

(3) KCl ($\alpha = 1$)

(4) $Al_2(SO_4)_3$ ($\alpha = 1$)

31) Two liquids A and B on mixing show positive deviation from Raoult's law. If the boiling temperature of pure A and B are 340K and 390K respectively then the boiling temperature of resulting solution may be :

(1) 393 K

(2) 342 K

(3) 386 K

(4) 337 K

32) Consider the following statements.
(A) Components of Azeotropic binary mixture can not be seperated by fractional distillation.
(B) Ethanol-water azeotropic mixture contains approximately 95% by volume of ethanol.
(C) A solution containing components A and B follows Raoult's law if A-B attraction force remain same as A-A and B-B attraction forces.
The correct statements are :-

(1) A and B only

(2) B and C only

(3) A, B and C

(4) A and C only

33) If molarity of a dilute solution is doubled, the value of molal elevation constant ($K_b$) will be ?

(1) Halved

(2) Tripled

(3) Unchanged

(4) Doubled

34)

The correct order of the wavelength of light absorbed by the following complexes is,
A. $[Co(NH_3)_6]^{3+}$ B. $[Co(CN)_6]^{3-}$
C. $[Cu(H_2O)_4]^{2+}$ D. $[Ti(H_2O)_6]^{3+}$
Choose the **correct** answer from the options given below:

(1) B < D < A < C

(2) B < A < D < C

(3) C < D < A < B

(4) C < A < D < B

35)

Which of the following are paramagnetic?
A. $[NiCl_4]^{2-}$ B. $Ni(CO)_4$
C. $[Ni(CN)_4]^{2-}$ D. $[Ni(H_2O)_6]^{2+}$
E. $Ni(PPh_3)_4$
Choose the **correct** answer from the options given below :

(1) A and C only

(2) B and E only

(3) A and D only

(4) A, D and E only

36) Which of the given statements are true :-
(1) In presence of ligand, crystal field splitting does not occur hence substance is colourless.
(2) The 'g' subscript is used for octahedral and square planar complex which have centre of symmetry
(3) If a spherically symmetrical field of negative charges surrounds the metal atom/ion the degeneracy is maintained.
(4) Toxic metal ion is removed by chelating ligand.

(1) FTTT

(2) TFTT

(3) TTFT

(4) TTTF

37) Choose correct order of C–O bond length :-

(1) $[Mn(CO)_6]^+ > [Cr(CO)_6] > [Ti\ (CO)_6]^{2-}$

(2) $[Cr(CO)_6] > [Mn(CO)_6]^+ > [Ti\ (CO)_6]^{2-}$

(3) $[Ti\ (CO)_6]^{2-} > [Cr(CO)_6] > [Mn(CO)_6]^+$

(4) $[Ti\ (CO)_6]^{2-} > [Mn(CO)_6]^+ > [Cr(CO)_6]$

38) $Ni^{+2}$ gives rosy red ppt. with dmg in basic medium. Which of the following option is incorrect regarding ppt ?

(1) Total number of chelate ring = 4

(2) Hybridisation is $dsp^2$

(3) It has symmetrical hydrogen bond

(4) It show's G.I. as well as O.I.

39)

The crystal field splitting energy for $CoCl_6^{-4}$ is 27000 $cm^{-1}$. The Δt for $[CoCl_4]^{-2}$ would be :-

(1) 143 kJ $mol^{-1}$

(2) 12000 $cm^{-1}$

(3) 16000 $cm^{-1}$

(4) both (1) and (2)

40) Identify the correct statement about $[Ni(H_2O)_6]^{+2}$ and $[Cu(H_2O)_6]^{2+}$ :

(1) All Ni–O and Cu–O bond lengths of individual species are equal.

(2) Ni–O (equatorial) and Cu–O (equatorial) bond lengths are shorter than Ni–O (axial) and Cu–O (axial) ones respectively

(3) All Ni–O bond lengths are equal whereas Cu–O (equatorial) bonds are shorter than Cu–O (axial) bond

(4) All Cu–O bond lengths are equal where as Ni–O (equatorial) bonds are shorter than Ni–O (axial) bonds.

41) Correct statement is.
(A) Complex $[Pt(NH_3)_2Cl_2]$ show geometrical isomerism.
(B) Optical isomers are mirror images that can not be superimposed on one another. These are called enantiomers.
(C) Chlorophyll is a complex compound of magnesium.
(D) Ethylenediamine is a mono dentate ligand.

(1) (A), (B), (C), (D) are correct options.

(2) (A), (B), (C) are correct options.

(3) (A), (B), (D) are correct options.

(4) (A), (D) are correct options.

42) The number of geometrical isomer found in complexes :-

$[PtCl_2(NH_3)_2]$, $[Ni(CO)_4]$, $[Ru(H_2O)_3Cl_3]$ and $[CoCl_2(NH_3)_4]^+$ respectively are :-

(1) 1,1,1,1
(2) 2,1,2,2
(3) 2,0,2,2
(4) 2,1,2,1

43) Which complex can show maximum types of isomerism ?

(1) $[Co(NO_2)(NH_3)_5]Cl_2$
(2) $[Co(NO_2)_2(NH_3)_4]Cl_2$
(3) $[Co(en)(NH_3)_4]Cl_3$
(4) $[Co(NO_2)_2(en)_2]Cl$

44) **Assertion :** $[CoCl_3(NH_3)_3]$ does not give white precipitate with $AgNO_3$ solution.
**Reason :** $[CoCl_3(NH_3)_3]$ complex is optically inactive.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

45) Which of following is not considered as organometallic compound ?

(1) Ferrocene
(2) Ziese's salt
(3) Grignard reagent
(4) Cis-platin

## BIOLOGY

1) **Assertion :** The codon for methionine and tryptophan are said to be non-degenerate.
**Reason :** There is single type of codon for both these type of amino acids.

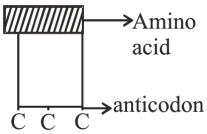
(1) If both assertion & reason are true & the reason is a **correct** explanation of the assertion.
(2) If both assertion & reason are true but reason is **not** a correct explanation of the assertion.
(3) If assertion is **true** but the reason is **false**
(4) If both assertion & reason are **false**

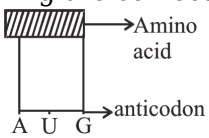
2) **Assertion (A) :** Mutation is a phenomenon which results in alteration of DNA sequences.
**Reason (R) :** Recombination and mutation leads to variation in DNA.

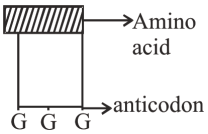
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.

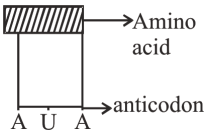
(4) Both Assertion & Reason are False.

3) Find out the sequence of binding of the following amino acyl-t-RNA complexes during translation of a m-RNA transcribed by a DNA segment having the base sequence. 3' ATACCCATGGGG 5'. Choose the answer showing the correct order of alphabets.

(a) 


(b) 


(c) 


(d) 


(1) a, b, c, d

(2) d, a, b, c

(3) a, b, d, c

(4) b, a, c, d

4) Which among these are salient features of human genome ?
(i) Human genome contains 3164.7 million bps.
(ii) Repeated sequence make up very small portion of human genome.
(iii) Chromosome one has 2968 genes.
(iv) Chromosome Y has 231 genes.
(v) More than 2 percent of the genome codes for proteins.

(1) i, ii, iii, iv

(2) i, ii, iv, v

(3) i, iii, iv

(4) i, ii, iii, iv, v

5) Match column-I to column-II.

| **Codon** | | **Amino acid** | |
|---|---|---|---|
| A | AAA | i | Glycine |
| B | CCC | ii | Lysine |
| C | UUU | iii | Phenylalanine |
| D | GGG | iv | Proline |
| | | v | Tyrosin |

(1) A-ii, B-v, C-iii, D-iv

(2) A-v, B-iii, C-i, D-iv

(3) A-ii, B-iv, C-iii, D-i

(4) A-iii, B-ii, C-v, D-i

6) If haploidal chromosome number for a diploid cell is 10, then monosomic number for this cell shall be :-

(1) 9

(2) 18

(3) 10

(4) 19

7)

During translation, peptide bond formation takes place between :

(1) –COOH of $(n + 1)^{th}$ amino acid and $–NH_2$ of $n^{th}$ amino acid

(2) $–NH_2$ of $n^{th}$ amino acid and –COOH of (n + 1) amino acid

(3) –COOH of nth amino acid and $–NH_2$ of $(n + 1)^{th}$ amino acid

(4) $–NH_2$ of nth amino acid and $–NH_2$ of $(n + 1)^{th}$ amino acid

8) DNA fragments generated by the restriction endonucleases in a chemical reaction can be separated by :

(1) Restriction mapping

(2) Centrifugation

(3) Polymerase chain reaction

(4) Gel-electrophorosis

9) Which one of the following does not follow the central dogma of molecular biology ?

(1) *E.coli*

(2) Retro-viruses

(3) Pea

(4) *Vicia faba*

10) Which of the following is not a level of gene expression regulation in eukaryotes?

(1) Transcriptional level (regulation of formation of primary transcript)

(2) Processing level (regulation of splicing)

(3) Transport of mRNA from nucleus to the cytoplasm

(4) Replication level (regulation of DNA synthesis)

11) How many statements are correct?
(a) VNTR belong to class of a minisatellite.
(b) Satellite DNA do not code for proteins.
(c) DNA from every cell or tissue from an individual show different degree of polymorphism.
(d) Number of repeats in VNTRs show very high degree of polymorphism.

(1) All statements

(2) Only one statement

(3) 2 statements

(4) 3 statements

12) The "cri-du-chat" syndrome is caused by change in chromosome structure involving:

(1) Deletion
(2) Duplication
(3) Inversion
(4) Translocation

13)

The given sequence when translated must form a polypeptide chain of how many amino acids?

5'-CUCAAAUCCAUGCCCCUGGGCUUGUAAGCGA-3'
(1) Eight
(2) Nine
(3) Six
(4) Five

14) Find the wrongly matched pair :

(1) Hargobind Khorana – Synthesis of RNA molecule with defined combination of bases
(2) George Gamow – Triplet Codon
(3) Meselson and stahl – Regulation of gene expression
(4) Alec Jeffreys – DNA Fingerprinting

15) Consider the following processes: Splicing, Replication, Transcription, Translation, Aminoacylation of tRNA. How many processes occur in nucleus?

(1) 3
(2) 4
(3) 5
(4) 2

16) Which of the following is a ribozyme in bacteria for the formation of peptide bond?

(1) 28 S rRNA
(2) 5.8 S rRNA
(3) 23 S rRNA
(4) 16 rRNA

17) One type of codon codes for only one type of amino acid. This property of genetic code/codon is known as :

(1) Non-ambiguous
(2) Degenerate
(3) Universal
(4) Comma less

18) What should be nature of codon if 6 type of N-bases, make code for 30 types of amino acids?

(1) Singlet
(2) Doublet
(3) Triplet
(4) Tetraplet

19) In lac operon, y gene codes an enzyme which is responsible for:

(1) Hydrolysis of lactose
(2) Increasing permeability of the cell to lactose (β-galacto side)
(3) Transfer of acetyl group
(4) Repression of lac operon

20) How many ATP and GTP are required for formation of polypeptide chain of 200 amino acids?

(1) 100 ATP, 200 GTP
(2) 200 ATP, 400 GTP
(3) 300 ATP, 400 GTP
(4) 1 ATP, 2 GTP

21) Movement of a gene from one chromosome to another non-homologous chromosome is known as:

(1) Duplication
(2) Translocation
(3) Crossing over
(4) Inversion

22) For DNA fingerprinting, DNA is not obtained from:

(1) WBC
(2) Hair root cell
(3) Skin cell
(4) RBC

23) Anticodon present on tRNA, complementary to initiator codon (AUG) of mRNA is :

(1) 3'CAU5'
(2) 3'GUA5'
(3) 3'TAC5'
(4) 3'UAC5'

24)

Find the correct statement about tobacco smoking–
(A) Nicotine inhibits secretion of adrenal medulla.

(B) Smoking is associated with increased incidence of cancer of lungs and throat.
(C)Smoking increases carbon monoxide content in blood and decreases the concentration of haembound oxygen.
(D) Smoking decreases blood pressure.

(1) A and D
(2) B and D
(3) A and B
(4) B and C

25) Viral DNA after being converted from viral RNA by enzyme X, incorporates into host genome to undergo replication. What is 'X'?

(1) DNA protease
(2) Restriction endonuclease
(3) RNA polymerase
(4) Reverse transcriptase

26) Which of the following may be caused due to alcoholism :
(A) Emphysema
(B) Amnesia
(C) Bronchitis
(D) Hypoglycemia
(E) Psychosis
Choose the correct option from given :-

(1) only D and E
(2) B, D, E
(3) Only B and D
(4) A, B, D, E

27) Which of the following statement is **false** ?

(1) Gametocytes, (sexual stages) of plasmodium develop in human RBC
(2) Plasmodium reproduce asexually in human liver cells.
(3) Fertilisation of plasmodium gametes take place in salivary gland of mosquito.
(4) Sporozoite is the infective stage for humans.

28) **Assertion :** Immune response, provided by antibodies is called as humoral immune response.
**Reason** : Antibodies are found in blood.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

29) Given below are two statement :

**Statement-I :** The foetus receives some antibodies from their mother through the placenta during pregnancy.
**Statement-II :** Receiving of antibodies through colostrum and placenta of foetus/newly born are some example of passive immunity.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are incorrect.

30) **Statement A :** Metastasis is the most feared property of benign tumor.
**Statement B :** Normal cells show a property called contact inhibition.

(1) Statement A is correct, Statement B is incorrect
(2) Statement A is incorrect, Statement B is correct
(3) Both are correct
(4) Both are incorrect

31) Given below are two statements –
**Statement-I :** HIV is a member of a group of viruses called Arbovirus.
**Statement-II :** Benign tumors normally remain confined to their original location.
In the light of the above statements choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are incorrect
(2) Statement-I is correct but Statement-II is incorrect
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and Statement-II are correct

32) **Statement-I :** Health simply means "absence of disease" or "physical fitness".
**Statement-II :** Health is not affected by genetic disorders, infections and life style.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

33) **Statement-I :** Pathogens that enter gut must know a way of surviving in stomach at low PH and digestive enzymes.
**Statement-II :** Injecting the microbes during vaccination induce active immunity.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

34) Match the columns with regards to Vector and Disease

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| p | *Culex* | i | Dengue |
| q | *Anopheles* | ii | Filariasis |
| r | *Aedes* | iii | Malaria |

(1) p-i, q-ii, r-iii

(2) p-ii, q-iii, r-i

(3) p-ii, q-i, r-iii

(4) p-i, q-iii, r-ii

35) Match the item of column-I with column-II and choose the correct options :-

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Nicotine | (i) | *Papaver somniferum* |
| (B) | Marijuana | (ii) | *Cannabis sativa* |
| (C) | Coca alkaloids | (iii) | *Erythroxylum coca* |
| (D) | Morphine | (iv) | *Nicotiana tabacum* |

(1) A-iv, B-ii, C-i, D-iii

(2) A-iv, B-iii, C-ii, D-i

(3) A-iv, B-ii, C-iii, D-i

(4) A-iv, B-i, C-iii, D-ii

36) Match column I with column II

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (i) | BCG | (a) | Passive immunity |
| (ii) | Polio | (b) | Toxoid |
| (iii) | Tetanus | (c) | Inactivated vaccine |
| (iv) | ADS | (d) | Attenuated vaccine |

(1) i-d, ii-c, iii-b, iv-a

(2) i-c, ii-d, iii-a, iv-b

(3) i-b, ii-a, iii-d, iv-c

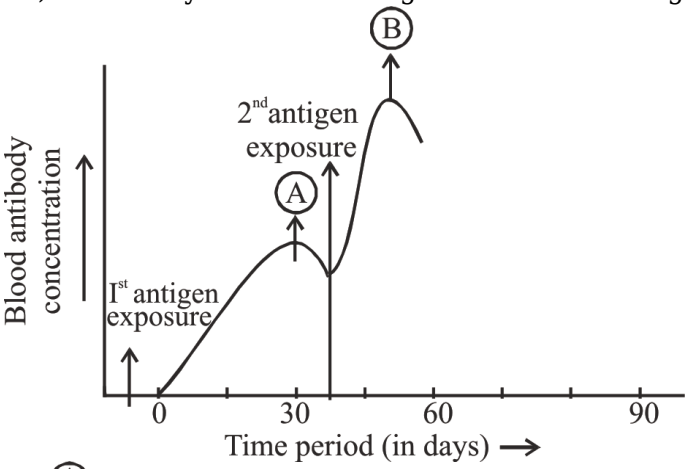
(4) i-a, ii-c, iii-b, iv-d

37) Reena is a 12 years old girl studying in higher secondary. Over the last few days, she was feeling tired. By the time, the weekend had arrived, she developed fever with chills, cough, headache and difficulty in breathing. On Sunday evening, her mother took her to doctor. On completing the examination, the doctor found out that her alveoli was filled with fluid causing difficulty in breathing.

The most probable diagnosis for the condition of Reena would be -

(1) Pneumonia
(2) Diphtheria
(3) Typhoid
(4) Amoebiasis

38) How many of the following are correct about given diagram ?




(a) Ⓐ is anamnestic response
(b) Memory cells are responsible for Ⓐ
(c) Memory cells are responsible for Ⓑ
(d) IgM is responsible for Ⓐ

(1) One
(2) Two
(3) Three
(4) Four

39)

Identify correct about allergy
(I) IgE are produced
(II) Anaphylatic shock is a allergic condition.
(III) Allergy treatment done by antihistamine.

(IV) AIDS is a allergic disorder.
(1) Only two statements are correct
(2) Only IV option is correct
(3) Only three statements are correct
(4) All correct

40) After immuno fluorescence technique done on stomach section, antibodies found in parietal cells are marker of :-

(1) Pernicious Anaemia
(2) Hashimoto's disease

(3) Rheumatoid Arthritis

(4) Thalassemia

41) Each molecule has four polypeptide chains ,two small called light chains and two longer called heavy chains (represented as $H_2L_2$). Which of the following is not directly related to above described molecule :-

(1) Army of proteins produced by B- lymphocytes

(2) Present in blood

(3) Humoral immunity

(4) Cell mediated immunity(CMI)

42) Which of the following is incorrectly matched ?

| | **Disease** | **Pathogen** | **Causative agent** |
|---|---|---|---|
| (1) | Penumonia | Hemophilus influenzae | Bacteria |
| (2) | Typhoid | Salmonella | Bacteria |
| (3) | Filariasis | Microsporum | Fungi |
| (4) | Chikungunya | Chikungunya virus | Virus |

(1) 1

(2) 2

(3) 3

(4) 4

43) Hashish and Ganja are got from :

(1) Erythroxylum

(2) Heroin

(3) Papaver

(4) Cannabis

44) Innate immunity has how many barriers?

(1) One

(2) Two

(3) Three

(4) Four

45) HIV that causes AIDS, first starts destroying :

(1) Leucocytes

(2) Helper T- Lymphocytes

(3) Thrombocytes

(4) B- Lymphocytes

46) Toxoid is used in vaccination against

(1) Tetanus
(2) Botulism
(3) Diphtheria
(4) All of these

47) Match the Following
Match the components with their roles in clotting
A. Platelets
B. Calcium ions
C. Prothrombin
D. Fibrin
i. Meshwork forming protein
ii. Cation acting as clotting factor
iii. Converts to thrombin
iv. Releases factors for clot formation
Option :-

(1) A-iv, B-ii, C-iii, D-i
(2) A-ii, B-iv, C-i, D-iii
(3) A-iii, B-i, C-iv, D-ii
(4) A-i, B-iii, C-ii, D-iv

48) Correct sequence in clot formation:
A. Prothrombin → Thrombin
B. Platelet plug formation
C. Fibrinogen → Fibrin
D. Vasoconstriction

(1) D → B → A → C
(2) B → D → C → A
(3) D → A → B → C
(4) A → C → B → D

49) Which organ receives blood through the hepatic portal vein?

(1) Heart
(2) Kidney
(3) Liver
(4) Brain

50) The hepatic portal system begins and ends with

(1) Veins
(2) Capillaries

(3) Arteries
(4) Arterioles

51) Which of the following is not a lymphatic organ?

(1) Spleen
(2) Thymus
(3) Liver
(4) Tonsil

52) Which valve prevents backflow of blood from left ventricle to left atrium?

(1) Tricuspid valve
(2) Bicuspid (mitral) valve
(3) Aortic valve
(4) Pulmonary valve

53) Aldosterone mainly increases reabsorption of

(1) Potassium
(2) Sodium
(3) Glucose
(4) Urea

54) Which cells are responsible for producing antibodies

(1) T-lymphocytes
(2) B-lymphocytes
(3) Monocytes
(4) Eosinophils

55) Which of the following is essential for the conversion of fibrinogen to fibrin during blood clotting?

(1) Thrombin
(2) Heparin
(3) Prothrombin only
(4) Platelets only

56) **Assertion (A):** A person with blood group AB is considered a universal recipient.
**Reason (R):** Their plasma lacks anti-A and anti-B antibodies.

(1) A and R both true and R correctly explains A
(2) A and R both true but R does not explain A
(3) A true, R false
(4) A false, R true

57) Which type of nephron has a long loop of Henle extending deep into the medulla?

(1) Cortical nephron
(2) Juxtamedullary nephron
(3) Renal corpuscle nephron
(4) Transitional nephron

58) **Assertion (A):** PCT reabsorbs maximum water from the filtrate.
**Reason (R):** PCT has abundant microvilli that increase the surface area for reabsorption.

(1) A and R are true, R explains A
(2) A and R are true, but R does not explain A
(3) A true, R false
(4) A false, R true

59) Afferent arteriole → Glomerulus → Efferent arteriole → Peritubular capillaries
What does this flow chart mainly depict?

(1) Lymph filtration
(2) Renal blood circulation
(3) Hormone secretion
(4) Venous return to heart

60) The Juxta-Glomerular Apparatus (JGA) helps regulate kidney function mainly by:

(1) Increasing water reabsorption in the collecting duct
(2) Releasing renin to regulate glomerular filtration rate
(3) Preventing ultrafiltration in the glomerulus
(4) Controlling urea reabsorption in the loop of Henle

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 2 | 4 | 4 | 4 | 1 | 1 | 2 | 1 | 2 | 1 | 2 | 3 | 4 | 2 | 2 | 2 | 1 | 1 | 2 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 3 | 2 | 1 | 3 | 3 | 4 | 3 | 3 | 2 | 3 | 3 | 3 | 2 | 2 | 1 | 1 | 2 | 4 | 4 | 3 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 2 | 3 | 1 | 2 | 2 | | | | | | | | | | | | | | | |

## CHEMISTRY

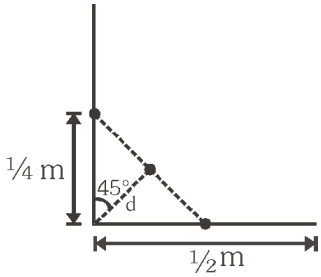
| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 4 | 2 | 4 | 4 | 1 | 3 | 2 | 3 | 1 | 3 | 1 | 3 | 4 | 3 | 1 | 2 | 2 | 2 | 1 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 4 | 3 | 3 | 2 | 2 | 3 | 4 | 3 | 2 | 2 | 4 | 3 | 3 | 2 | 3 | 1 | 3 | 4 | 4 | 3 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 2 | 3 | 4 | 2 | 4 | | | | | | | | | | | | | | | |

## BIOLOGY

| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 1 | 2 | 2 | 3 | 3 | 4 | 3 | 4 | 2 | 4 | 4 | 1 | 4 | 3 | 1 | 3 | 1 | 2 | 2 | 2 |
| Q. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| A. | 2 | 4 | 4 | 4 | 4 | 2 | 3 | 1 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 1 | 1 | 2 | 3 | 1 |
| Q. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 4 | 3 | 4 | 4 | 2 | 4 | 1 | 1 | 3 | 2 | 3 | 2 | 2 | 2 | 1 | 1 | 2 | 1 | 2 | 2 |

SOLUTIONS

PHYSICS

1) 


$d = \frac{1}{4}\sin 45^\circ = \frac{1}{4\sqrt{2}}m$

d = 0.1767 m or 17.7 cm

2) $X_{CM} = \frac{m\left(\frac{\ell}{2}\right) + m\left(\frac{\ell}{2}+\frac{\ell}{2}\right) + m\left(\frac{\ell}{2}+\frac{\ell}{2}+\frac{\ell}{4}\right) + m\left(\frac{\ell}{2}+\frac{\ell}{4}+\frac{\ell}{6}+\frac{\ell}{2}\right)}{m+m+m+m}$

$X_{CM} = \frac{25\ell}{24}$

3)

Net external force is in vertically downward direction.

4) $50 \times 10 = 1000 \times v \quad \square \quad v = \frac{1}{2}$ m/s

$E_i = \frac{1}{2} \times \frac{50}{1000} \times 10 \times 10 = 2.5J$

$E_f = \frac{1}{2} \times \frac{1000}{1000} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}J$

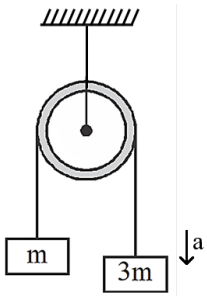
$\%\ \text{loss} = \frac{2.5 - 1/8}{2.5} \times 100 = 95\%$

5) Here impulse only depende on change is momentum of ball is
20–(–10) = 30 NS

6) $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) u_2$ and

$v_2 = \left(\frac{2m_1}{m_1 + m_2}\right) u_1 + \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_2$

on putting the values $v_1$ = 6 m/s and $v_2$ = 12 m/s

7) 


$a = \frac{3mg - mg}{4m} = \frac{g}{2}$

$= \frac{3ma - ma}{4m}$ down wards

$\frac{a}{2} = \frac{g}{4}$ down wards

8) $3m(200\cos 60°)\hat{i} = (m \times 100)\hat{j} + (m \times 100)\left(-\hat{j}\right) + m\vec{v} \quad \vec{v} = \left(300\,\hat{i}\right)$ m/s

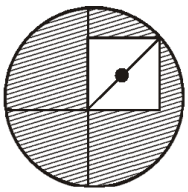
300 m/s in the horizontal direction.

9) $\frac{M}{2}x = M(10 - x)$

$\Rightarrow x = \frac{20}{3}m$

10) $v_1 = \sqrt{2gh_1}$, $v_2\sqrt{2gh_2}$

$\Delta P = m\,(\vec{v}_2 - \vec{v}_1) = m(\sqrt{2gh_1} + \sqrt{2gh_2})$



11) mass $\propto$ area

$X_{CM} = \frac{M_1x_1 - M_2x_2}{M_1 - M_2}$

$X_{CM} = \frac{0 - 1/2r^2(r/2)(\sigma)}{\sigma\pi r^2 - r^2/2\,(\sigma)}$

$\Rightarrow X_{CM} = \frac{-r}{4\pi - 2}$

12) Net force on the system is in vertical direction.

13) $\vec{F}_{ert} = 1 \times 1 + 2 \times 2 - 4 \times .5 = 3N$

14) No horizontal thrust, velocity of trolley remain constant and horizontal momentum conserved.

15) According to Newton's third law they will always apply equal and opposite force on each other.

16) $\mu mg = \sqrt{(ma_t)^2 + (ma_c)^2}$

$\mu g = \sqrt{a_t^2 + a_c^2}$ [$\because a_t = \alpha r$]

$25 = \left(\frac{1}{\sqrt{2}}\right)^2 + a_c^2$

$\frac{49}{2} = a_c^2$

$a_c = \frac{7}{\sqrt{2}} = \omega^2(1)$

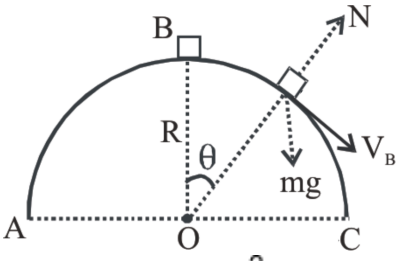
$\frac{7}{\sqrt{2}} = 2\alpha\theta = 2\left(\frac{1}{\sqrt{2}}\right)\theta \Rightarrow \theta = \frac{7}{2}$

$2\pi N = \frac{7}{2}$

$N = \frac{7}{4\pi}$

17)

Let the car looses the contact at angle θ with vertical




$mg\cos\theta - N = \frac{mv^2}{R} \Rightarrow N = mg\cos\theta - \frac{mv^2}{R}$

During descending on overbridge θ is increase.
So cosθ is decrease therefore normal reaction is decrease.

18) During a circular turn, friction provides the necessary centripital force

$f = \frac{mv^2}{R}$ and also $f \leqslant f_L$

So, (A) and (R) are both correct and (R) correctly explanation (A)

19)

$\theta = \omega t$

$\theta = \omega \times \frac{\pi}{2\omega}$

$\theta = \frac{\pi}{2}$

Both the particles will rotate by 90°.

$\vec{V}_1 = -\theta R_1 \hat{i}$

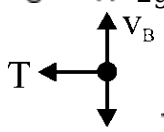
$\vec{V}_2 = -\omega R_2 \hat{j}$

$\square\ \vec{V}_1 - \vec{V}_2 = -\omega R_1 \hat{i} + \omega R_2 \hat{j}$

20) **For (A)** $v_B^2 = v_A^2 - 2gR = 9gR - 2gR = 7gR \Rightarrow v_B = \sqrt{7gR}$

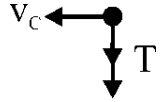
**For (B)** $v_C^2 = v_A^2 - 2g(2R) = 5gR \Rightarrow v_C = \sqrt{5gR}$

**For (C)**




$T_B = \frac{mv_B^2}{R} = 7mg$

**For (D)**




$T_C + mg = \frac{mv_C^2}{R} \Rightarrow T_C = 4mg$

21)

Given, $a_c = k^2rt^2 \Rightarrow \frac{v^2}{r} = k^2rt^2$

$\Rightarrow V = krt$

and $a_T = \frac{dv}{dt} = kr$

Tangential force $F_T = ma_T = mkr$

Power imparted by $F_T = F_T V = (mkr)(krt)$

$= mk^2r^2t$

Power imported by $f_c = 0$

Net power $P = mk^2r^2t$

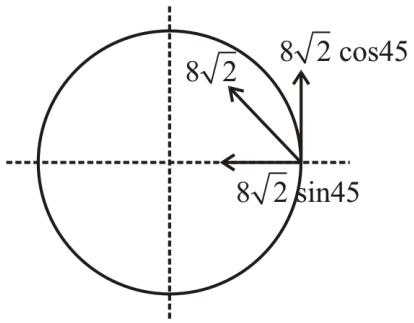
22)

Let x be increase in length of the spring. The particle would move in a circular path of radius ($\square$ + x).

Centripetal force = force due to the spring

$m(\square + x)\ \omega^2 = kx$

$\therefore x = \frac{m\omega^2\ell}{k - m\omega^2}$




23)

$a_c = 8\sqrt{2}\sin 45 = 8 m/sec^2$

$a_c = \frac{v^2}{r} = 8$

$r^2 = 8r = 8 \times 2 = 16$

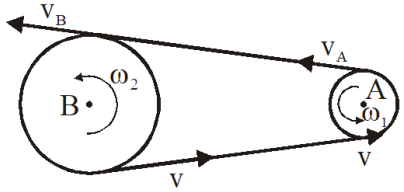
v = 4m/sec

24) $\vec{r} = \cos\omega t\hat{i} + \sin\omega t\hat{j}$
$\vec{v} = -\omega\sin\omega t\hat{i} + \omega\cos\omega t\hat{j}$
$\vec{a} = -\omega^2\cos\omega t\hat{i} + \omega^2\sin\omega t\hat{j}$
$\Rightarrow \vec{v}.\vec{r} = 0$
velocity is perpendicular to $\vec{r}$. $\Rightarrow \vec{a} = -\omega^2(\vec{r})$

25) $\theta = \theta_0 + \theta_1 t + \theta_2 t^2$
$\frac{d\theta}{dt} = \omega = 0 + \theta_1 + \theta_2(2t)$
$\frac{d\omega}{dt} = \alpha = 0 + \theta_2(2)$
$\alpha = 2\theta_2$ Independent of time, so a straight line parallel to y-axis
$a_t = R\alpha$
$a_t = R(2\theta_2)$
$a_t$ = constant (Independent of time)

26) Acceleration of bob is towards centre of circle O. Hence resultant force is also towards O.

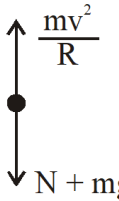
27) tangential vel. are same




$v_B = v_A \Rightarrow \omega_2 R_2 = \omega_1 R_1 \Rightarrow \frac{\theta_1}{\theta_2} = \frac{\omega_1}{\omega_2} = \frac{R_2}{R_1} = 2$

28) $v_{max} = \sqrt{\mu_s rg} = 8.0$ m/s

29) $mg(h - 2R) = \frac{1}{2}mv^2$ ...(i)




$N + mg = \frac{mv^2}{R}$ ; as $N = 3mg \Rightarrow 4g = \frac{v^2}{R}$..(ii)
Solving equation (i) & (ii) h = 4R

30) Acceleration = $\frac{v^2}{R}$

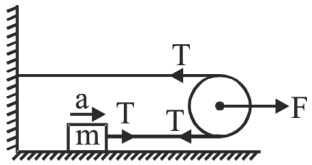
31) Change in momentum,

$\Delta p = \int F dt$

= Area of F-t graph

$= \frac{1}{2} \times 2 \times 6 - 3 \times 2 + 4 \times 3$

= 12 N-s

32) 


Block : T = ma

F = 2T ⇒ T = F/2

$a = \frac{F}{2m}$

If the block moves by a distance d, pulley moves by d/2

$a_p = \frac{a}{2} = \frac{F}{4m}$

33) Since block is in motion, friction will be

μmg = 60 N

Hence option (2)

34) friction will act opposite to the relative velocity.

35) Let T is the tension of the rope. When monkey climbs up with acceleration a, then

$T - m_1 g = m_1 a$ ....(i)

When another monkey climbs down with same acceleration a, then

$m_2 g - T = m_2 a$ ....(ii)

Hence, $(m_2 - m_1)g = (m_1 + m_2)a$

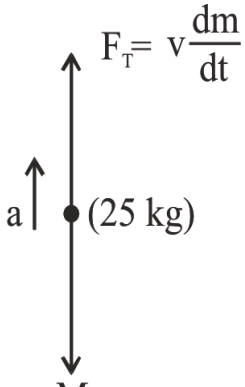
or $\left(1 - \frac{m_1}{m_2}\right) g = \left(\frac{m_1}{m_2} + 1\right) a$

or $\left(1 - \frac{2}{3}\right) g = \left(\frac{2}{3} + 1\right) a$

or $a = \frac{g}{5}$

36) $a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g = 6 m/sec^2$ $a_{rel} = 12 m/s^2$

$S_{rel} = \frac{1}{2} a_{rel} t^2 \Rightarrow 6 = \frac{1}{2} \times 12 \times t^2$ ⇒ t = 1



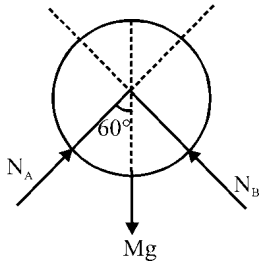

37)

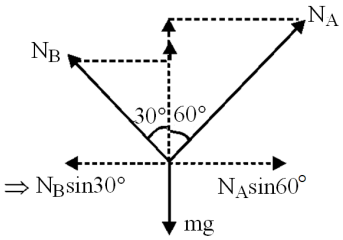
$F_T - Mg = Ma$

$F_T = M(g + a)$

$v\frac{dm}{dt} = 25(9.8 + 9.8)$

$\frac{dm}{dt} = \frac{25(2 \times 9.8)}{49 \times 10^2} = 0.1(kg/s)$




38)

$N_A \sin 60° = N_B \sin 30°$

$N_A \cdot \frac{\sqrt{3}}{2} = N_B \frac{1}{2}$

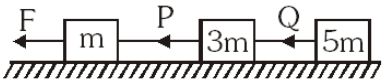
$\sqrt{3}N_A = N_B$

$\boxed{\sqrt{3}N_A = N_B}$

39) For equilibrium :- $f \geq mg$

$\mu Kd \geq mg \quad K \geq \frac{mg}{\mu d}$

Hence $K_{min} = \frac{mg}{\mu d}$




40)

Let a be the common acceleration of the system

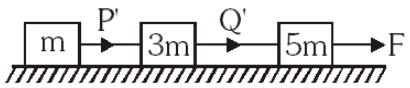
$\Rightarrow a = \frac{F}{m + 3m + 5m} = \frac{F}{9m}$ ....(i)

$\Rightarrow P = (3m + 5m)a$

$16 = 8ma \quad [\Rightarrow P = 16N(Given)]$

$a = \frac{2}{m}$ ....(ii)

Substituting this value of a in eqn(i), we get F = 18 N




P' = ma = 2N ( a = 2/m)
Q' = (3m + m)a = 4ma = 4 × 2 N = 8 N.

41) $F = \frac{mdv}{dt}$

42) Net force acts towards earth on the satellite.

43) Area of a–x graph

$= \int a\,dx = \int \left(v\frac{dv}{dx}\right) dx = \int v\,dv$

$= \int_0^{v_m} v\,dv = \frac{1}{2} \times 1 \times 1$

$\Rightarrow \frac{v_m^2}{2} = \frac{1}{2} \Rightarrow v_m = 1$ m/s

44) **For (1) :** $a = \frac{50}{1+1} = 25 ms^{-2}$, contact force between A and B = 1 × 25 = 25N
**For (2) :** For system, reaction given by ground = 20 + 30 = 50N
For block A reaction between A & B = 50 – 10 = 40N
**For (3) :** For motion of A, $N_{AB}$ – 10 = 1 × 25 Þ $N_{AB}$ = 35N
**For (4) :** For motion of A, $N_{AB}$ = 1 × 35 = 35N

45) Acceleration of a system = $\frac{\text{Net force in the direction of acceleration}}{\text{total mass of the system}}$

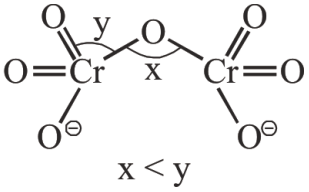
$\Rightarrow a = \frac{10g\sin 53^\circ - 4g\sin 37^\circ - 4g}{10+4+2+4} = 0.8\ m/s^2$

CHEMISTRY

46)

NCERT-XII, Pg # 223 & 236

47) **Solution**: A, B, C are correct facts about the dichromate/chromate system. D. Incorrect: In the dichromate ion, the Cr-O-Cr bond angle (≈126º) is larger than the O-Cr-O bond angle (≈109.5º) in the tetrahedral units.




x < y

x = 126°

**Final Answer**: Option (4)

48) **Solution**:

A. Colorless ions (P) →$d_0$ ($Sc^{3+}$) or $d_{10}$ ($Zn^{2+}$). →(iv)
B. Coloured ions (R) →$d_{1-9}$ ($Fe^{3+}$,$Cu^{2+}$). →(i)
C. μs=5.9 BM (Q) →n=5 ($Fe^{3+}$,$Mn^{2+}$). →(iii)
D. μs=1.7 BM (S) →n=1 ($Ti^{3+}$,$Cu^{2+}$). →(ii)

**Final Answer**: (P)→(iv),Q→(iii),(R)→(i),S→(ii).

49)

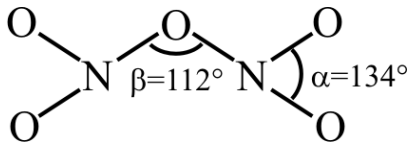
$KMnO_4$ preparation involves $MnO_2$ oxidation in an alkaline medium. Options list $MnO_2$ with KOH/NaOH and oxidizing agents. $KNO_3$, $O_2$, and $KCO_3$ all serve as effective oxidizing agents. Each option provides the necessary reactants for the reaction. Therefore, all listed conditions can be used for $KMnO_4$ preparation. Hence, "All of these" is the correct answer.

**Final Answer** - Option (4)

50) **Question Explanation**: Identifying the type of carbide not formed when lanthanides are heated with carbon.
**Concept**: Chemical Reactivity of Lanthanides.
**Solution**: Lanthanides (Ln) react with carbon at high temperatures (2773 K) to form various carbides. The common formulas are Ln3C (monocarbides), $Ln_2C_3$ (sesquicarbides), and $LnC_2$ (dicarbides). The formula $Ln_2C_2$ is not typically formed or is an alternative way to write the dicarbide structure ($Ln_2(C_2)$), but $Ln_2C_2$ as a separate entity is not a conventionally listed carbide type. **Calculation**: No calculation is required. **Final Answer**: $Ln_2C_2$.

51)

Conceptual

52) 


53) **Question Explanation:** Count molecules producing dibasic acids.
**Concept:** Hydrolysis of halides and oxides.
**Solution:**
$PCl_5 + HOH \rightarrow H_3PO_4 + HCl$
(Basicity = 3)
$SF_6 + HOH \rightarrow$ No hydrolysis
$SF_4 + HOH \rightarrow H_2SO_3 + HF$
(Basicity = 2)
$P_4O_6 + HOH \rightarrow H_3PO_3$

(Basicity = 2)

$PCl_3 + HOH \rightarrow H_3PO_3 + HCl$

(Basicity = 2)

$SF_4$, $P_4O_6$, and $PCl_3$ produce acids with a basicity of 2 upon hydrolysis.

**Final Answer:** (3) 2

54)

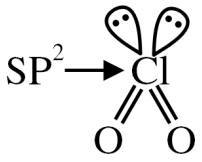
NCERT-XII, Part-I Page No. # 188

55)

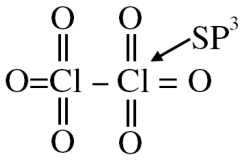
NCERT-XII, Page No. # 209, Part-I, Edition-2019

56)

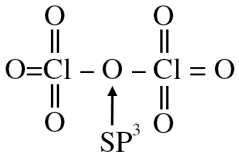
$ClO_2$, $Cl_2O_6$ , $Cl_2O_7$, $Cl_2O$
are N.M.O. → Acidic in nature



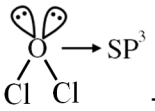

→ Paramagnetic on hydrolysis gives → $HClO_2$ & $HClO_3$




→ on hydrolysis gives → $HClO_3$ & $HClO_4$




→ on hydrolysis gives → $HClO_4$

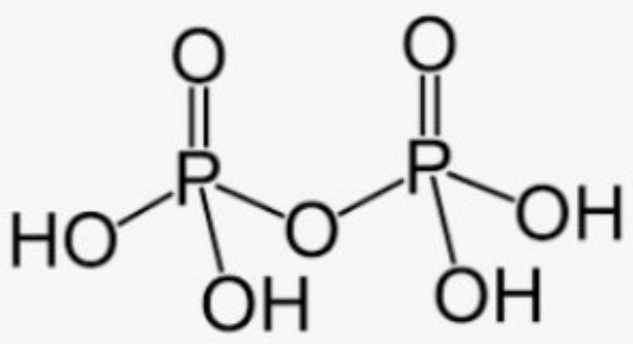



→ on hydrolysis gives → HClO

57)

**Explanation –** The question is asking about the characteristic types of bonds and their numbers in the structure of pyrophosphoric acid. Pyrophosphoric acid has the formula ($H_2P_2O_7$).

**Concept – oxoacids of phosphorus**
**(Bonding P-OH , P = O & P-O-P in $H_2P_2O_7$)**




Four P–OH, Two P = O, One P–O–P
The correct answer is option (1)

58) Rhombic Sulphur and monoclinic Sulphur have different crystal structure

59)

Sn decompose steam to form dioxide and $H_2$ gas

60)

Conceptual

61) **Question Explanation**
Question is asking about the compound which is soluble in hot water.
**Concept**
This Concept based on Solubility of compound in water **Solution**
The solubility of $PbCl_2$ increases with temperature because the increased kinetic energy of water molecules at higher temperatures helps to dissolve it. So the compound lead(II) chloride ($PbCl_2$) is soluble in water.
**Final Answer** (1)

62)

$CaO + H_2O \rightarrow Ca(OH)_2$
(A) less soluble
$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$
(B) insoluble
$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$
B Soluble

63) **Question Explanation:**
Question is asking about result if HCI is not added before passing $H_2S$ in the second group.

**Concept:**
This question is based on Group analysis, Group reagent

**Solution:**
If HCI is not added before passing $H_2S$ in the second group, it may result in the precipitation of Group IV cations
This causes confusion and incorrect group identification.

**Hence,**
**Correct Option: 2**

64)

**Explaining the Question :-**

Question is asking about incorrect match of identification test and colour

**Concept:-**

identification test and colour

**Solution :-**

**Chromyl chloride ($CrCl_3$)** is **red or pink** in color instead of green hence this match is incorrect
**Final Answer :- 2**

65)

**Question Explanation :**
Question is asking about assertion and reason type question.

**Concept :**
This question is based on Group analysis, cations.

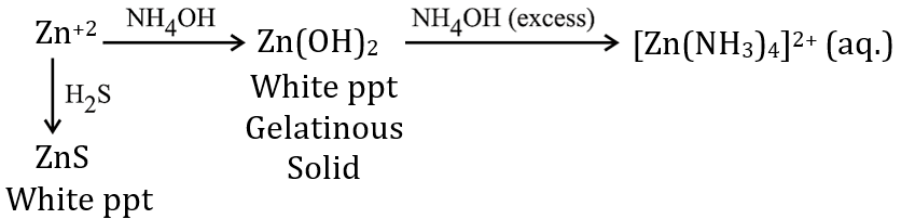
**Solution :**

Ammonium ions ($NH_4^+$) are typically identified $i_z$ a separate "zero" group analysis because they are distinguished by their unique properties and do not react with the group regents used $i_z$ subsequent group. So to not intergere $i_z$ the test they are positioned in seperate "zero group". So, assertion and reason are true and reason is correct explanation of assertion.

**Hence,**

**Option (1) is correct :** Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

66) **Explaination:** Question is asking about $M^{+2}$ ion.
**Concept:** Colour of gas, colour of Precipitate.
**Solution:**




**Final Answer:** Option (4)

67) $Y_A = \frac{X_A P_A^o}{X_A P_A^o + X_B P_B^o}$

$= \frac{1}{1 + \frac{X_B P_B^o}{X_A P_A^o}} = \frac{1}{1 + \frac{2}{1} \times \frac{2}{1}} = \frac{1}{5} = 0.2$

68)

$i = 1 + (n - 1)\alpha = 1 + 2 \times 0.6 = 2.2$
$\Delta T_f = iK_f m = 2.2 \times 1.86 \times 0.2 = 0.82$
$T_f = 0 - 0.82 = -0.82°C$

69)

$M = \frac{M \times 1000}{1000 \times d - M \times M_w \text{of solute}}$

$= \frac{2 \times 1000}{1000 \times 1.8 - 2 \times 98}$

$= 1.25$

70)

$\Delta T_b = iK_b m$
$\Delta T_b \propto im \Rightarrow T_b \propto im$

71) $K_b = \frac{R(T_b^0)^2}{1000 \times L_v}$

$L_v$ - latent heat of vapourization

72)

PQ - curve of VP of solvent
XY - curve of VP of solution
$\Delta T$ - elevation in BP

73)

$\pi = iCRT$

(a) 0.001 M $Na_2SO_4 \Rightarrow \pi_1 = 3(0.001)$ RT
0.003 M Glucose $\Rightarrow \pi_2 = 1(0.003)$ RT
$\pi_1 = \pi_2$ (isotonic)
(b) 0.003 M $Na_2SO_4 \Rightarrow \pi_1 = 3(0.003)$ RT
0.003 M Glucose $\Rightarrow \pi_2 = 1(0.003)$ RT
$\pi_1 > \pi_2$ (Hypertonic)
(c) 0.001 M $Na_2SO_4 \Rightarrow \pi_1 = 3(0.001)$ RT
0.01 M Glucose $\Rightarrow \pi_2 = 1(0.01)$ RT
$\pi_1 < \pi_2$ (Hypotonic)
(d) Osmotic pressure
$\pi = iCRT$

74)

For ideal solution
$\Delta H_{mix} = 0$ $\Delta V_{mix} = 0$
$\Delta S_{mix} > 0$ $\Delta G_{mix} < 0$

75)

$i = 1 + (n - 1)\alpha$
For $AlCl_3$ ($\alpha = 1$) i = 4
For $CaCl_2$ ($\alpha = 1$) i = 3
For KCl ($\alpha = 1$) i = 2
For $Al_2(SO_4)_3$ ($\alpha = 1$) i = 5

76)

Solution show positive deviation has BP point less than both the component.

77) NCERT-XII (2017), Pg. # 45 **OR** For ideal solution force of attraction does not change an mixing.

78)

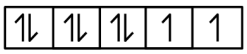
$K_b$ depends on nature of solvent.

79)

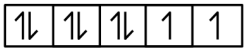
Strength of ligand $\uparrow$ $\Delta_0 \uparrow$
Absorbed Energy $\uparrow$ $\lambda_{absorbed} \downarrow$
**OR**
The 'g' subscript (g-gerade meaning even) is symmetry label used in point groups that posses a center of symmetry

80)

(A) $[NiCl_4]^{2-}$ ⇒ Oxidation state = +2
Electronic configuration = $3d^8\ 4s^0$
Cl = W.F.L. No Pairing

 2 unpaired electrons

(D) $[Ni(H_2O)_6]^{2+}$
Oxidation state = +2, configuration = $3d^8$
$H_2O$ is W.F.L. No pairing

 2 unpaired electrons

Hence both are paramagnetic
**OR**
Crystal field splitting only occurs when the field has a specific geometry.

81)

Chelation therapy helps remove these toxic metals from the body by forming stable, soluble complexes.

82) Synergic bonding Metal – carbon B.O. ↑
C–O B.O. ↓

83) $[Ni(dmg)_2]$ $dsp^2$

84) $\Delta_0(CoCl_6)^{-4} = 27000\ cm^{-1}$
for tetrahedral complexes:

$$\Delta t = \frac{4}{9}\Delta_0$$

$$\Delta t = \frac{4}{9} \times 27000 = 12000\ cm^{-1}$$ options (2)

convert 12000 $cm^{-1}$ → kJ $mol^{-1}$ using
1 $cm^{-1}$ = 0.01196 kJ $mol^{-1}$
12000 × 0.01196 = 143.5 kJ $mol^{-1}$

85) If the d electrons are unsymmetrically arranged, they will repel some ligands in the complex more than others. Thus the structure is distorted.

86) NCERT-XII, Pg. No. # : 245

87)

$[Pt\ NH_3\ Cl_2]$ → $dsp^2$ → 2G.I.
$[Ni(CO)_4]$ → $sp^3$ → 0 G.I.
$[Ru(H_2O)_3Cl_3]$ → fac, Mer 2 G.I.
$[COCl_2\ (NH_3)_4]$ → 2 G.I.

88) $[Co(NO_2)_2(en)_2]Cl$ can show ionisation, linkage, geometrical and optical isomerism.

89)

**Explanation -** The question asks to evaluate the truth of an assertion and reason about $[CoCl_3(NH_3)_3]$ and determine if the reason explains the assertion.
**Concept -** (reaction with $AgNO_3$, G.I. & O.I. in Complex)

A. The assertion is true: $[CoCl_3(NH_3)_3]$ ] has all Cl- inside, so it doesn't precipitate AgCl with $AgNO_3$.
B. The reason is true: both fac and mer isomers of $[CoCl_3(NH_3)_3]$ are optically inactive due to planes of symmetry.
C. The reason, optical inactivity, does not explain why the complex doesn't react with $AgNO_3$.
D. Therefore, both are true, but the reason is not the correct explanation of the assertion.

**Final Answer - Option (2)**

90)

Fact based

BIOLOGY

91)

NCERT XII, Pg # 96

92) NCERT Pg. # 86

93)

Codon and anticodon should be antiparallel complementary to each other for pairing.

94)

NCERT XII, Pg # 104

95)

NCERT XII, Pg # 96

96)

For diploid cell, monosomic $n\theta = (2n - 1)$

97)

Refer to module

98)

Refer to chapter 'biotechnology principles and processes".

99)

Riboviruses show RNA replication and retroviruses show transcription.

100)

NCERT XII, Pg. # 99

101) NCERT Pg. # 121-122, para 6.10

102)

Refer to module

103)

I$^{st}$ amino acid - due to start codon (AUG)
Last amino acid due to stop codon (UAA)

104) NCERT-XII, Pg. # 95, 89, 106

105)

NCERT Pg # 114

106)

NCERT XII, Pg. # 99

107)

NCERT XII, Pg. # 96

108)

Refer to module

109)

NCERT XII, Pg. # 101

110)

Refer to module

111)

Refer to module

112)

Mature RBCs do not have nucleus and hence DNA.

113)

Codon and anticodon should be antiparallel and complementary to each other for pairing.

114) **Explanation:**

A. (B) Smoking is indeed associated with an increased incidence of cancer of the lungs and throat.
B. (C) Smoking increases the carbon monoxide content in the blood, which decreases the concentration of hemoglobin-bound oxygen because carbon monoxide binds to hemoglobin more effectively than oxygen.

**The correct statement about tobacco smoking is: Option 4: B and C**

115)

NCERT–XII, Pg. # 139

116)

NCERT–XII, Pg. # 146

117)

NCERT–XII, Pg. # 132

118) NCERT-XII Pg#148
Antibodies are found in blood, so antibody mediated immunity is called as humoral immunity.
प्रतिरक्षी, रक्त में पायी जाती है इसलिये प्रतिरक्षी माध्यित प्रतिरक्षा, तरल प्रतिरक्षा कहलाती है।

119) NCERT Pg # 152

120)

**Explain Question :** The question is asking to evaluate two statements about tumors and normal cells.
**Concept :** This question is based on cancer.
**Solution :**
Statement A : **Incorrect** – Metastasis is the most feared property of malignant tumors, not benign.
Statement B : **Correct** – Normal cells stop dividing when they come in contact with neighboring cells due to contact inhibition.
**Final answer :** 2

121) NCERT-XII Pg. # 154, 157

122) NCERT Pg. No. # 145, 146/ 158, 159

123) NCERT Pg. No. # 152/166

124) **Explaining the Question:**
Question is asking about matching of vector with related disease.

**Concept:**
This question is based on Bacterial disease, Protozoan disease, Helminthic disease.

**Solution:**

A. **(p) Culex:** mosquitoes are the primary vectors for **(ii) Filariasis.**
B. **(q) Anopheles:** Mosquitoes are the vectors for **(iii) Malaria.**
C. **(r) Aedes:** Mosquitoes are the vector for **(i) dengue.**

So, correct matching is: p-ii, q-iii, r-i

**Hence,**
**Option (2) is correct : p-ii, q-iii, r-i**

125)

NCERT Pg. # 142

126) NECER-XII, Pg # 152

127)

NCERT–XII, Pg. # 131

128)

NCERT–XII, Pg. # 135

129) NCERT–XII, Pg. # 153, Para-8.2.5

130)

NCERT–XII, Pg. # 137

131) **Explanation:**
The molecule described ($H_2L_2$) is an **antibody,** also known as immunoglobulin. Antibodies are produced by B lymphocytes and are a key component of the **humoral immune response.**

**Correct Answer: 4.** Cell-mediated immunity (CMI)

132)

NCERT–XII, Pg. # 133

133) NCERT XII$^{th}$ Pg.#159

134) NCERT Pg. # 150 Para 8.2.1

135)

NCERT–XII, Pg. # 139

136)

NCERT–XII, Pg. # 136

137) “platelets initiate, fibrin finishes.”

138) Vessel tightens, platelets gather, enzymes activate, fibrin seals.

139) This vein carries nutrient-rich blood from intestines to the liver.

140) It connects two sets of capillaries — gut and liver.

141) Liver belongs to the digestive system, not lymphatic.

142) Bi = two; present on the left side of the heart.

143) Acts on DCT & collecting duct.

144) Think “B” for “Builder” of antibodies.

145) The active enzyme that knits fibrin threads.

146) No antibodies means no attack.

147) Long loops help in producing concentrated urine.

148) Microvilli-rich PCT is suited for maximum reabsorption.

149) Shows the special capillary network of the nephron.

150) JGA → renin → regulates GFR via RAAS.